

The Concrete Facts

Q: What is autoclaved aerated concrete (AAC)? Why have I only heard about it recently, when apparently it has been around for years? When should a builder such as myself decide to use this material?

— Wondering in Winnipeg

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MARY JAVES

A: AAC is a manufactured building product that can be made in a large variety of sizes and shapes. It is a pre-cast masonry form, and is typically made from natural materials. It provides an easily assembled building surface that reduces material use (particularly for exotic or uncommon materials), lowers embodied energy, reduces cooling energy use, and can provide solidification of fly ash.

AAC Basics

AAC is made by mixing lime, sand (silica), water, and Portland cement with an aeration or expansion agent (typically aluminum powder). This slurry mixture is poured into a mold, and the alkali compounds in the mix react with the expansion agent for several hours until the mixture takes the form of a light-colored dried cake. For structural integrity, reinforcement in the form of steel bars or mesh can be added to the mold during the first stage of production. The aluminum—while less than 10% of the product—is critical; its reaction with the concrete causes microscopic bubbles to form in the mix, greatly expanding the volume of the concrete. For this reason, AAC is sometimes referred to as autoclaved cellular concrete.

The speed of gas evolution, the consistency of the slurry, and the

slurry's setting time all must be appropriate to one another. Fly ash, a waste product from coal-fired power plants, can be added to the mixture as a substitute for some of the sand (although it will cause the color of the AAC to become darker and grayer).

Once the moisture and hydrogen gas have evaporated, the concrete form can be cut to either a standard or a custom size and then cured in a steam-pressurized chamber (the autoclave). Typically, conditions in the chamber are in the range of 375°F and 175 lb per square inch (psi). The final product, a hardened calcium silicate, is airtight, nontoxic, and—depending on whether or not it contains steel reinforcing bars or steel mesh—can be used either in load-bearing or non-load-bearing walls, or in floors, roofs, or lintels.

Advantage to Builders

Building professionals value the characteristics and utility of AAC. First, the material's density is usually 25–45 lb/ft³, but it can range from as low as 15 lb/ft³ up to 60 lb/ft³. Common AAC has less than 20% to about 35% of the density of standard hardened concrete. Because of its lower weight, AAC is much easier to handle and install than is concrete.

Second, this lower density and structural simplicity make it easy for contractors to saw, drill, rout, or nail AAC in desired applications. The air pockets in the material are tiny enough that the material will be unlikely to chip or crumble.

Third, because of these tiny, noninterconnecting air pockets, AAC has insulating abilities (R-values up to 1.25 per inch) that are several times those of concrete. Thus, its energy efficiency can approach or exceed that of a standard insulated stud wall. Tests performed over the last ten years at the Fraunhofer Institute for Architectural Physics in Germany—which does research in building acoustics, heat technology, and new building materials—showed that under outside temperatures of more than 120°F, the inside of a 10-inch-thick AAC wall maintained tempera-

tures of 66°F–70°F over a 24-hour period. Also, a significant delay or lag time from maximum interior temperature occurs several hours after the exterior temperature reaches its peak. (For more details on the thermal effects of massive walls, see "Mass Walls Mean Thermal Comfort," *HE* May/June, '99, p. 37.) Thus, energy consumption, while not substantially reduced, is shifted to off-peak hours—a benefit both to power companies and to customers who want to avoid blackouts. Note, however, that this characteristic is exhibited in warm climates; in colder winters, when temperatures can stay cold for weeks, that thermal lag makes little difference. The increased R-value, thermal mass, and airtightness working together give AAC good acoustical properties, reducing noise infiltration.

Finally, the product is noncombustible, so it holds a fire rating that exceeds all existing fire-rating codes. Fire ratings for AAC are typically around one hour of delay per inch of AAC thickness. Although I have not confirmed this with regional insurance companies, AAC may help to reduce fire insurance premiums with some companies.

One drawback of AAC is that, because of its lower density, it has a



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Masons find AAC an easy material to use.

compressive strength of just 300–1,000 psi, whereas standard concrete can be 2–3 tons per square inch. Thus, standard AAC is fine for buildings up to two stories high, but for taller structures it must be reinforced. Therefore, builders

constructing multifamily homes and apartments would use AAC with steel reinforcement, which is more expensive. For builders constructing single-family homes, from entry-level models to custom projects, AAC can also be used as simple veneer or in more creative masonry designs. Patterns, reveals, signage, and graphics can be created on AAC surfaces, and AAC can be carved or sculpted for bas-relief.

Uses

AAC was developed in the early twentieth century in Sweden, and was commercially produced starting around 1930. By the early 1990s, more than 50 factories worldwide were producing more than 31 million cubic meters of AAC blocks and panels; these totals had increased to more than 50 million cubic meters in at least 150 plants by 1998. Most of this manufacturing capacity is in Europe, but the market has recently expanded in Asia, and it has spread to Australia, South America, and the United States.

In the United States, the technology has not been used extensively, due to the dominance of wood construction. Over the past seven years, however, three European construction companies have introduced AAC to the U.S. market. All three plants are in the southeast—Hebel opened a plant in Georgia in 1996; YTONG opened another in Florida in 1997; and Babb International began operations in Georgia in 1999. The current domestic market is limited to the Southeast, Midwest, and Northeast, but plans are currently under way for the construction of several other facilities in other regions in the country—including a plant in Arizona that will provide AAC to western markets this fall (see “Concrete Suggestions,” *HE* Jan/Feb ’01, p. 3).

Although builders in the United States have had limited experience with AAC so far, considerable interest in this material has sparked efforts on the part of technical committees of professional societies—such as the American Concrete Institute (ACI) and the American Society for Testing and Materials (ASTM)—to

produce documents that cover the design and use of AAC. ASTM Committee C27 (Precast Concrete) has a newly developed standard approved for publication entitled “Standard Specification for Precast Autoclaved Aerated Concrete Wall Construction Units,” and ACI Committee 523 (Cellular Concrete) is working to develop a document on the design of AAC.



AAC is cut on a band saw designed specifically for AAC.

Products and Installation

AAC can be used in many applications because it is available in several forms. The precast masonry blocks are typically 2 ft long, 8 inches high, and 3–12 inches wide; these blocks are not usually reinforced in manufacturing. Precast wall and roof panels are available in 2-ft widths, thicknesses of 3–16 inches, and can be cut at any length up to 20 feet. These are reinforced, as are AAC lintels.

AAC blocks can be laid by a mason very quickly, even in the 12 inch x 24 inch block size, because of their low weight. Common carpentry tools can be employed on these blocks and panels, but concrete saws and drills are often used for fasteners and tie-downs. Common mortar or thin-set glue mortar can be used to set the blocks or panels.

Because it has the consistency of a hardened sponge, and because it is friable, AAC must be protected from moisture. Exterior walls made of this material can be plastered or stuccoed, or they can be finished with veneer, vinyl siding, brick, or

vinyl acrylic paint. For interior walls, drywall, tile, plaster, or paint can be used. As long as the AAC is not exposed to moisture, it gradually loses the water content added during manufacture. The water content stabilizes at approximately 5% after one to two years.

Green Issues

As is the practice of this column, I will examine several issues that are key to green building: material use, energy and resource use, hazardous chemical use, and disposal—as well as the important factor of cost.

Material use. AAC not only uses common materials, it uses less of them than other precast construction products. Because it is made with an expansion agent, it is less dense than other insulating products; thus it has a lower material use than concrete, brick, plaster, or stucco. AAC is often made with fly ash, a waste material. In this case, its raw material content can be even lower than when it is made with sand. In addition, the use of precut and custom-cut blocks and panels reduces waste at the construction site.

Energy and resource use. The embodied energy of AAC (based on process energy requirements for basic material only) is significantly lower than that of metals, plastics, and engineered-wood board (that is, particle board, plywood, and medium-density fiber board). It is also lower than that of most solid timber—hardwoods and some, though not all, softwoods. Of common building materials, only brick, stone, and standard concrete have lower embodied energy than AAC. The reason, of course, is that AAC must be cured by autoclaving to obtain its useful qualities.

What about the energy-saving characteristics of AAC? Its insulating capabilities are good, because its high thermal mass provides a heat-transfer delay. The effective thermal performance of the wall would be better than that of an R-10 frame wall in a climate where outdoor temperatures cool down at night. Thus the thermal performance of AAC is dependent on



AAC offers design flexibility, as here with the wall and the window trim. Note the quoins and keystones.

the climate in which it is used; in northern winters (like those in Winnipeg), when temperatures can stay cold for weeks, a thermal lag like this may actually be a detriment because it takes so much energy to heat the building up. Finally, the reduced weight of AAC in relation to concrete lowers the shipping and handling costs, and thus the energy associated with transportation.

Hazardous chemicals. No hazardous or toxic materials are used in the manufacturing of AAC, with the exception of the portion of fly ash that is under 7 microns in diameter. Particles smaller than this size can cause both acute and chronic respiratory problems. However, this is only a problem during the manufacture of the blocks; once the concrete block or panel is made, it is inert—no toxic material or compound is released. Because it is inert, AAC does not off-gas and poses no indoor health problems.

Disposal. The material is very durable. There are AAC structures more than 50 years of age that are in excellent condition. Because AAC products are inert, they will not rot, warp, rust, corrode, or decompose. Thus AAC will not cause or initiate disposal problems, unless it becomes saturated and ruined. The product can be recycled into lower-grade products as long as fly ash is not an ingredient, because the fly ash will be released when the AAC is reprocessed; fly ash is classified as a hazardous material and its release is not allowed. AAC that does not

contain fly ash is technically recyclable, but as yet no politically or economically viable system exists to collect and reprocess it.

Costs. AAC costs are quite competitive with those of other, more common building materials. A completed AAC wall typically costs less than a completed masonry or wood-framed wall, although it could cost up to 5% more in regions with more lumber resources and less accessibility to AAC. Both insulation and energy costs are often lower than such costs for conventional construction, at least in warmer climates, due to the insulating value of AAC. Transportation costs are lower than for other concrete, brick, or stone materials because AAC weighs less. And labor costs at construction sites will also be reduced because walls, roofs, floors, and other building assemblies can be created more quickly with AAC than with other types of masonry materials. Installation of AAC has been running \$3–\$4/ft² in industrialized countries, including those parts of the United States in which AAC is now available.

Guidelines

Clearly, for environmental construction, AAC is one material to consider. Builders must be cognizant of climate limitations to the insulating potential of AAC. Such limitations do not preclude the use of AAC; they simply make it necessary to use additional materials, such as rigid foam insulation. Contractors need minimal experience to begin working with AAC, although some of the knowledge required to utilize this material in construction is needed at the design stage.

With the growing need to reduce consumption of primary resources as the world's population increases and consumption of natural resources continues to grow, alternative building products like AAC will probably become more prevalent. Contractors will need to be familiar with such products and will have to consider environmental factors when making construction and remodeling decisions. As the construction industry continues to adapt, AAC is one of a growing list of environmentally preferable products that will save money for consumers and provide some environmental benefits.



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